

❑ SUPPLY CHAIN & LOGISTICS ANALYTICS DASHBOARD

A Comprehensive Data Analytics Project using PostgreSQL & Power BI

Developed By

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Tools Used

PostgreSQL

SQL

Power BI

DAX

Power Query

Project Type: Personal Portfolio Project

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1. Project Overview

1.1 Introduction

The logistics industry generates large volumes of operational data every day, including information related to customers, deliveries, fleet operations, drivers, fuel consumption, and maintenance activities. Analyzing this data helps organizations monitor business performance, improve operational efficiency, reduce costs, and make informed decisions.

This project focuses on developing an interactive **Logistics Operations Dashboard** using SQL and Microsoft Power BI. The dashboard provides a comprehensive view of the company's performance through five analytical sections: Executive Overview, Customer & Route Performance, Fleet Management, Driver Operations, and Delivery & Safety Operations. Together, these dashboards enable stakeholders to monitor key performance indicators (KPIs), identify business trends, and support data-driven decision-making.

1.2 Problem Statement

Logistics companies often manage large amounts of operational data that are stored across multiple datasets. Without proper analysis, it becomes difficult to monitor business performance, identify operational inefficiencies, control costs, and make timely strategic decisions.

The objective of this project is to transform raw logistics data into meaningful business insights by building interactive dashboards that allow management to analyze revenue, customer performance, fleet operations, driver activities, and delivery performance in a single reporting solution.

1.3 Objectives

The main objectives of this project are:

- To analyze overall business performance using key financial and operational KPIs.
- To identify high value customers, profitable freight categories, and high performing transportation routes.
- To monitor fleet utilization, fuel expenditure, maintenance costs, and vehicle availability.
- To evaluate driver and delivery performance using operational metrics.
- To build interactive Power BI dashboards that support data driven business decisions.

1.4 Tools Used

The following tools were used throughout the project:

- **Neon Console (PostgreSQL)** - Cloud-based PostgreSQL database used for data storage, data cleaning, SQL query execution, and data transformation.
- **Microsoft Power BI** - Dashboard development, data modeling, DAX calculations, and interactive data visualization.
- **Microsoft Excel** - Data validation and preliminary data exploration.

1.5 Dataset Overview

The project uses a logistics operations database consisting of multiple related tables that capture different aspects of the company's operations.

The major datasets include:

- **Drivers** - Stores driver information, employment details, and license records.
- **Trucks** - Contains fleet information, including truck specifications, acquisition details, and operational status.
- **Trailers** - Maintains trailer inventory, trailer types, and availability status.
- **Customers** - Stores customer accounts, contract information, and customer details.
- **Facilities** - Contains information about terminals and warehouse locations.
- **Routes** - Includes origin destination pairs, travel distances, and route pricing details.
- **Loads** - Records shipment information, freight type, customer assignments, and revenue generated.
- **Trips** - Captures actual trip execution, including assigned drivers, trucks, trailers, trip duration, and fuel consumption.
- **Fuel Purchases** - Stores fuel transaction details such as purchase amount, fuel cost, and purchase location.
- **Maintenance Records** - Contains vehicle maintenance history, maintenance type, repair costs, and vehicle downtime.
- **Delivery Events** - Records pickup, delivery, detention time, and on-time delivery performance.
- **Safety Incidents** - Stores accident records, violations, and damage costs associated with trips.
- **Driver Monthly Metrics** - Provides monthly aggregated performance measures for each driver.
- **Truck Utilization Metrics** - Contains monthly utilization statistics for fleet vehicles.

These tables are connected through primary and foreign key relationships, enabling integrated analysis across customers, routes, fleet operations, deliveries, maintenance, fuel consumption, and driver performance. This relational structure supports comprehensive business analysis and the development of interactive dashboards for operational and strategic decision making.

2 – EXECUTIVE SUMMARY

2.1 Dashboard Overview

Dashboard Objective

The **Executive Summary Dashboard** provides a centralized overview of the organization's logistics and supply chain performance by integrating key financial, operational, fleet, and workforce metrics into a single interactive dashboard. It enables senior management to monitor overall business performance, evaluate operational efficiency, and identify critical trends without navigating multiple reports.

The dashboard combines executive-level Key Performance Indicators (KPIs) with analytical visualizations to present a comprehensive view of revenue generation, operational costs, fleet availability, maintenance expenditure, and driver workforce distribution. This consolidated reporting approach supports faster decision-making, improves operational transparency, and enhances strategic planning across the logistics network.

Business Problem

Large logistics organizations generate massive volumes of operational data from transportation, fleet maintenance, financial transactions, and workforce management. When this information is distributed across multiple reports and systems, executives face challenges in obtaining a complete understanding of organizational performance.

Without a centralized analytical dashboard, monitoring financial performance, evaluating fleet utilization, tracking maintenance expenses, and assessing workforce distribution becomes time-consuming and inefficient. This often delays strategic decision-making and reduces overall operational visibility.

The Executive Summary Dashboard addresses these challenges by integrating multiple business domains into a single interactive reporting interface, allowing management to evaluate organizational performance efficiently and make data-driven business decisions.

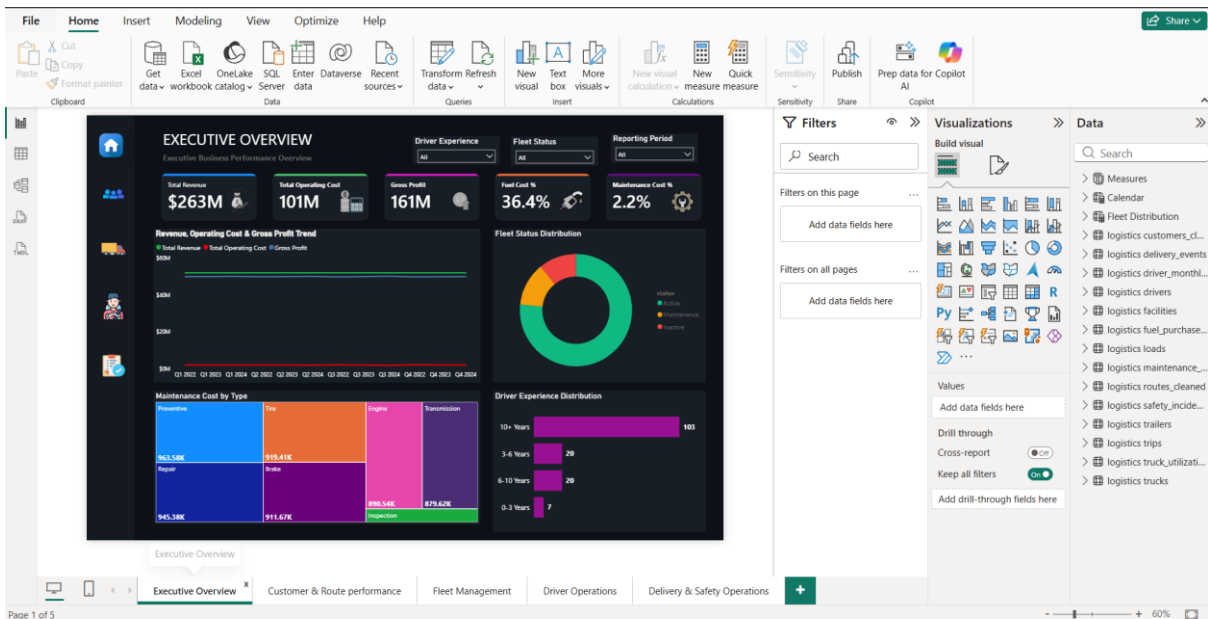
Dashboard Features

The Executive Summary dashboard combines high-level business metrics with interactive visualizations to provide a complete overview of organizational performance.

The key features include:

- Executive KPI cards summarizing financial and operational performance.
- Revenue, operating cost, and gross profit trend analysis.

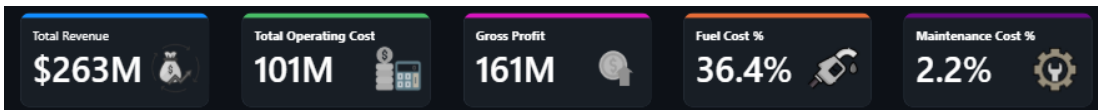
- Fleet status distribution monitoring.
- Maintenance cost analysis by maintenance category.
- Driver workforce experience distribution.
- Interactive filters for Driver Experience, Fleet Status, and Reporting Period.



2.2 Dashboard Components

Dashboard KPIs

The dashboard begins with five executive Key Performance Indicators (KPIs) that provide an immediate overview of the organization's financial performance, operational efficiency, and maintenance expenditure.



Executive KPI Summary Strip

1. Total Revenue

Value: \$263 Million

Description

Total Revenue represents the overall income generated from logistics operations during the reporting period. It serves as the primary financial performance indicator and reflects the organization's business growth and transportation demand.

Business Importance

- Measures business growth.
 - Evaluates organizational financial performance.
 - Supports strategic planning.
 - Assists executive decision-making.
-

2. Total Operating Cost

Value: \$101 Million

Description

Operating Cost represents the total expenses incurred while managing transportation, fleet operations, workforce activities, and logistics services. Monitoring this KPI helps management evaluate operational efficiency and cost control.

Business Importance

- Tracks operational expenditure.
 - Supports cost optimization.
 - Evaluates operational efficiency.
 - Improves financial planning.
-

3. Gross Profit

Value: \$161 Million

Description

Gross Profit measures the organization's profitability after deducting operating expenses from total revenue. It provides an overall assessment of financial health and operational sustainability.

Business Importance

- Measures business profitability.
- Supports investment decisions.
- Evaluates financial sustainability.
- Assists long-term planning.

4. Fuel Cost %

Value: 36.4%

Description

Fuel Cost Percentage indicates the proportion of operational expenditure allocated to fuel consumption. Since fuel is one of the largest transportation expenses, this KPI plays a critical role in cost management.

Business Importance

- Evaluates fuel efficiency.
- Supports fuel cost optimization.
- Monitors transportation expenses.
- Improves fleet operational planning.

5. Maintenance Cost %

Value: 2.2%

Description

Maintenance Cost Percentage represents the share of operational expenditure dedicated to vehicle maintenance activities. Continuous monitoring ensures vehicle reliability while minimizing unexpected repair costs.

Business Importance

- Measures maintenance efficiency.
- Supports preventive maintenance planning.
- Improves fleet reliability.
- Reduces unexpected breakdowns.

2.3 Dashboard Visual Analysis

1. Revenue, Operating Cost & Gross Profit Trend

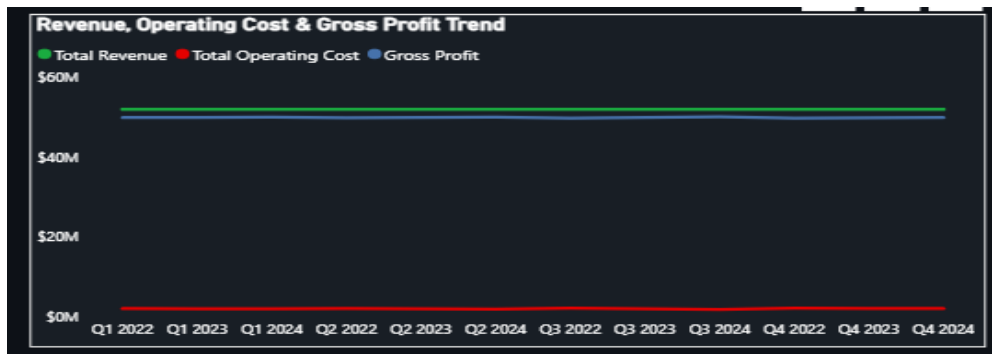


Figure 2.1: Revenue, Operating Cost and Gross Profit Trend

Description

This line chart illustrates the quarterly trend of Total Revenue, Operating Cost, and Gross Profit across the reporting period. It enables management to compare financial performance over time while monitoring the relationship between revenue generation and operational expenditure.

The visualization demonstrates that revenue and gross profit remain relatively stable throughout the reporting period, while operating costs are consistently controlled. This indicates balanced financial performance and effective operational cost management.

Business Importance

- Monitors financial performance over time.
- Identifies revenue growth patterns.
- Supports profitability analysis.
- Assists executive financial planning.

2. Fleet Status Distribution

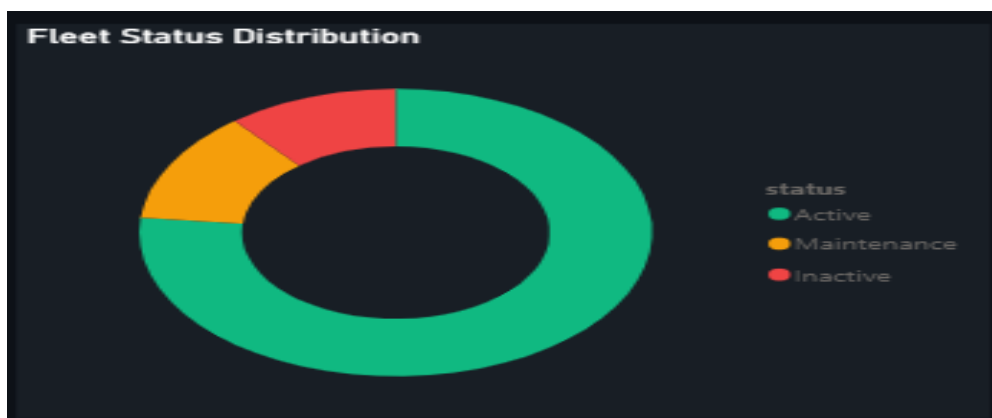


Figure 2.2: Fleet Status Distribution

Description

The Fleet Status Distribution donut chart categorizes fleet vehicles into **Active**, **Maintenance**, and **Inactive** groups. This visualization provides management with an overview of fleet availability and operational readiness.

A high proportion of active vehicles indicates efficient fleet utilization, while maintenance and inactive vehicles highlight opportunities for improving maintenance scheduling and asset utilization.

Business Importance

- Evaluates fleet availability.
 - Supports maintenance planning.
 - Improves vehicle utilization.
 - Reduces operational downtime.
-

3. Maintenance Cost by Type

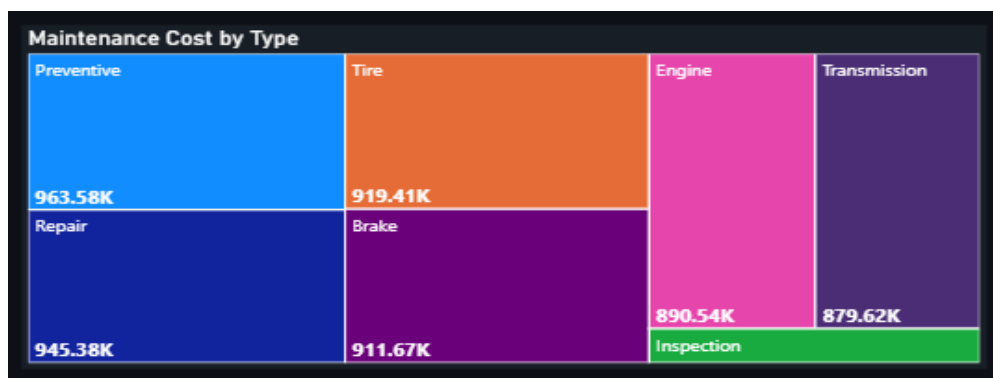


Figure 2.3: Maintenance Cost by Type

Description

The treemap displays maintenance expenditure across different maintenance categories, including Preventive Maintenance, Repairs, Tires, Brakes, Engines, Transmission, and Inspection.

The visualization helps management identify maintenance categories contributing the highest costs and supports budget allocation for fleet maintenance programs.

Business Importance

- Identifies high-cost maintenance activities.
- Supports preventive maintenance strategies.
- Optimizes maintenance budgeting.
- Improves fleet reliability.

• 4. Driver Experience Distribution

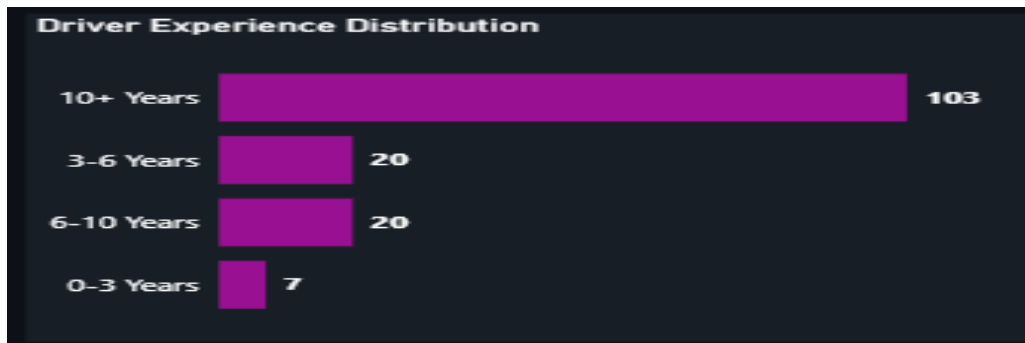


Figure 2.4: Driver Experience Distribution

Description

The Driver Experience Distribution bar chart classifies drivers according to their years of professional experience. The visualization provides insights into workforce composition and assists management in workforce planning and training initiatives.

The dashboard indicates that the majority of drivers possess more than ten years of experience, suggesting a highly experienced workforce capable of supporting efficient transportation operations.

Business Importance

- Evaluates workforce experience.
- Supports recruitment planning.
- Identifies training requirements.
- Improves workforce management.

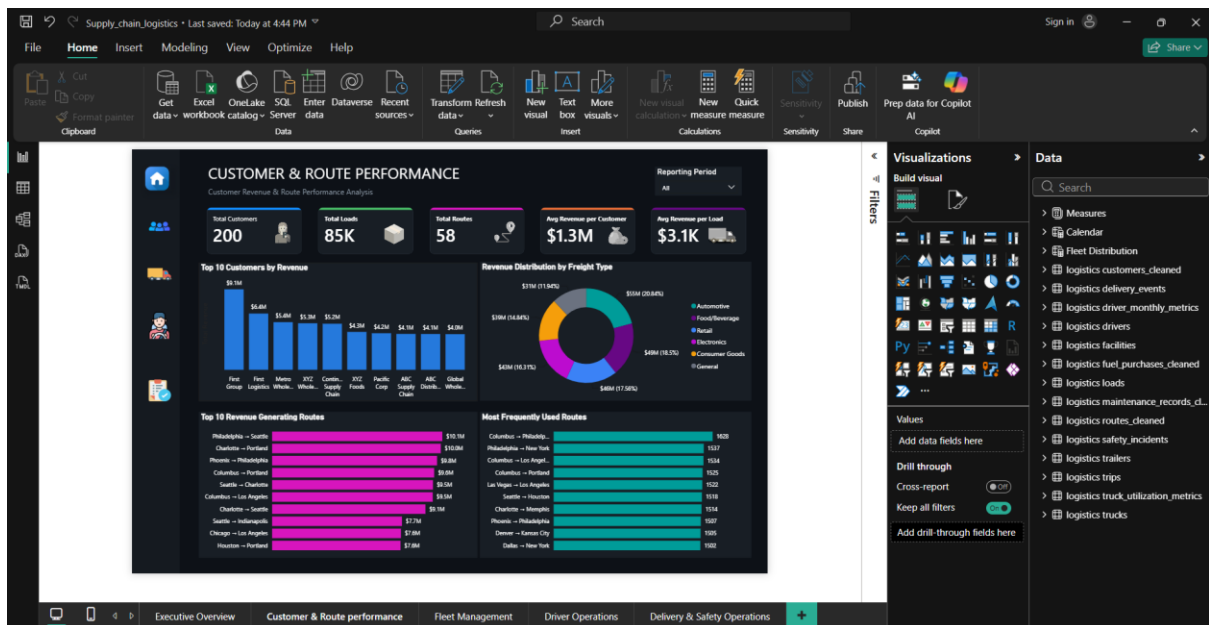
2.4 Executive Summary

The Executive Summary Dashboard successfully integrates financial performance, fleet operations, maintenance analytics, and workforce distribution into a single executive reporting platform. By combining high-level KPIs with interactive visualizations, the dashboard provides decision-makers with a comprehensive overview of organizational performance. The centralized reporting approach enhances operational visibility, supports strategic planning, and enables faster, data-driven business decisions across the logistics network.

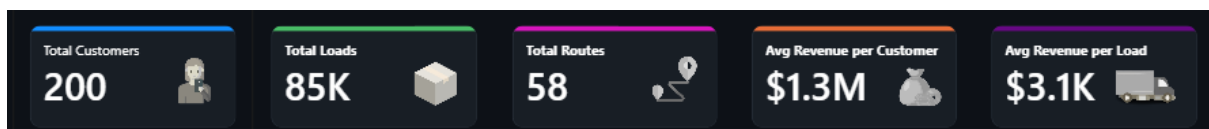
3. Customer & Route Performance

3.1 Dashboard Objective

The Customer & Route Performance Analysis dashboard evaluates the organization's revenue generation across customers, freight types, and transportation routes. It provides insights into customer contribution, route utilization, and freight distribution, enabling management to identify high-value customers, optimize transportation routes, and improve operational efficiency. The dashboard also includes a reporting period filter that allows users to analyze business performance across different quarters and years.



3.2 Key Performance Indicators (KPIs)



The Customer & Route Performance Analysis dashboard consists of five key performance indicators that provide an overview of customer engagement, transportation activity, and revenue performance. The company serves a total of **200 customers** and has successfully completed **85,000 freight loads** across **58 transportation routes**, reflecting the scale of its logistics operations.

The dashboard also highlights two revenue-based performance metrics. The **Average Revenue per Customer** is **\$1.3 million**, indicating the average revenue contribution of each customer to the business. Additionally, the **Average Revenue per Load** is **\$3,100**, providing insight into the average income generated from each completed freight load. Together, these KPIs help management evaluate customer value, operational productivity, and overall business performance.

3.3 Visual Analysis

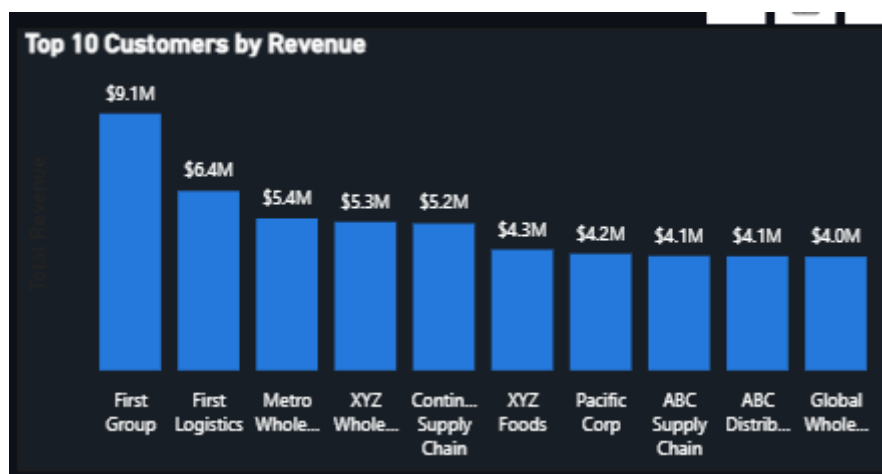
The Customer & Route Performance Analysis dashboard includes four analytical visualizations to evaluate customer contribution and route efficiency:

- **Top 10 Customers by Revenue:** A vertical bar chart displaying the ten highest revenue-generating customers, enabling management to identify key business accounts and their contribution to overall revenue.

SQL Query : The visualization was developed by joining the Customers and Loads tables using the customer_id field. Revenue was aggregated for each customer using the SUM(revenue) function, after which the results were sorted in descending order and the top ten customers were selected.

```
Page 2 ...  Save Primary Idle neondb
33
34 -- top 10 customers by revenue
35 select c.customer_id,
36        c.customer_name,
37        sum(l.revenue) as total_revenue
38 from logistics.loads l
39 join logistics.customers_cleaned c on c.customer_id = l.customer_id
40 group by c.customer_id, c.customer_name
41 order by total_revenue desc
42 limit 10;
43
```

Dashboard Visualization :

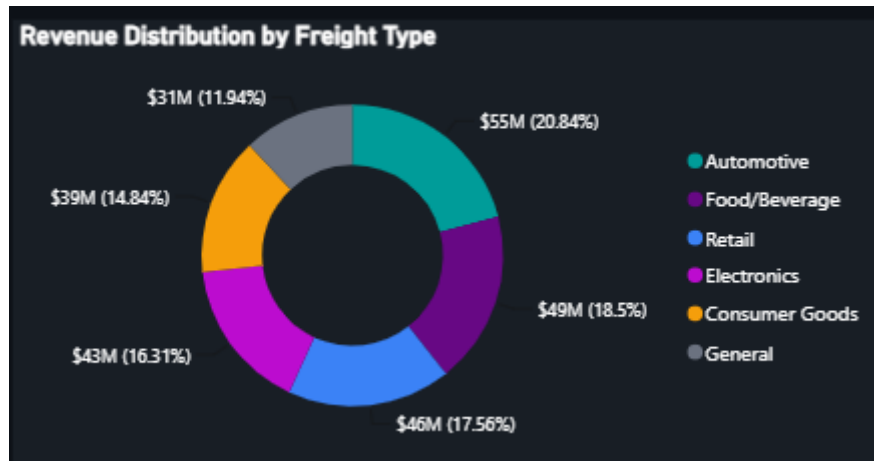


- **Revenue Distribution by Freight Type:** A donut chart illustrating the proportion of revenue generated by different freight categories, helping identify the company's primary revenue-generating services.

SQL Query : The chart was developed using the Loads and Customers table. Revenue was grouped by primary_freight_type and aggregated using the SUM(revenue) function. The total revenue for each freight category was then used to calculate the distribution of revenue across different freight types.

```
Page 2 ... Save Primary Active neondb
68 group by account_status,
69
70 -- freight type revenue
71 select c.primary_freight_type,
72 sum(l.revenue) as total_revenue
73 from logistics.loads l
74 join logistics.customers_cleaned c on c.customer_id = l.customer_id
75 group by c.primary_freight_type
76 order by total_revenue desc;
77
```

Dashboard Visualization :

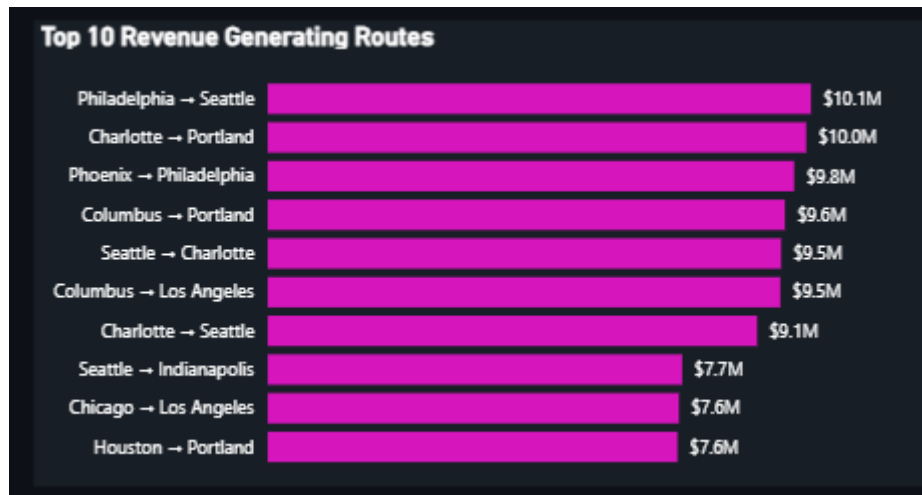


- **Top 10 Revenue-Generating Routes:** A horizontal bar chart highlighting the transportation routes that generate the highest revenue, supporting strategic route planning and resource allocation.

SQL Query : The visualization was created by joining the Routes and Loads tables using the route_id field. Revenue was aggregated for each route using the SUM(revenue) function. The routes were then ranked in descending order based on total revenue, and the top ten revenue-generating routes were displayed.

```
Page 2 ... Save Primary Active neondb
76 order by total_revenue desc;
77
78 -- Which top 10 routes generate the highest revenue
79 select r.route_id,
80 r.origin_city || ', ' || r.origin_state || ' → ' || r.destination_city || ', ' || r.destination_state AS route,
81 sum(l.revenue) as total_revenue
82 from logistics.routes_cleaned r
83 join logistics.loads l on l.route_id = r.route_id
84 group by r.route_id, r.origin_city, r.origin_state, r.destination_city, r.destination_state
85 order by total_revenue desc
86 limit 10;
87
```

Dashboard Visualization :



- **Most Frequently Used Routes:** A bar chart showing the routes with the highest number of completed loads, providing insight into route utilization and operational demand.

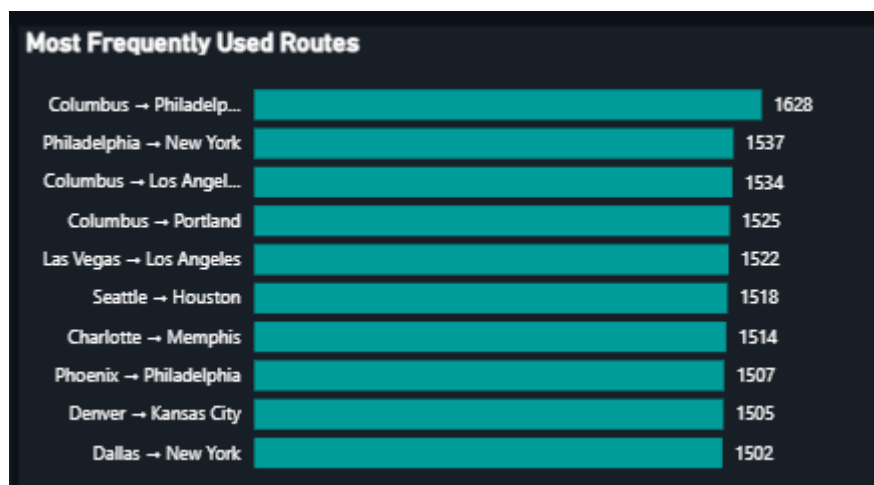
SQL Query : The chart was created by joining the Routes and Loads tables using the route_id field. The total number of freight loads was calculated for each route using the COUNT(load_id) function. The routes were then ranked based on shipment frequency, and the ten most frequently used routes were displayed.

```

87
88 -- Which routes are used most frequently?
89 select count(l.load_id) as total_loads,
90 r.origin_city || ', ' || r.origin_state || ' → ' || r.destination_city || ', ' || r.destination_state AS route
91 from logistics.routes_cleaned r
92 join logistics.loads l on l.route_id = r.route_id
93 group by r.route_id, r.origin_city, r.origin_state, r.destination_city, r.destination_state
94 order by total_loads desc
95 limit 10;
96

```

Dashboard Visualization :



- **Reporting Period Filter:** A Year and Quarter slicer has been included to enable users to analyze customer revenue, freight distribution, and route performance across different

reporting periods. This interactive filter allows management to compare business performance over time and identify seasonal or quarterly trends that support informed decision-making.

3.4 Business Insights

- **Insight 1: Revenue Generation is Concentrated Among High-Value Customers**

The Top 10 Customers by Revenue visualization reveals that a relatively small group of customers contributes a substantial share of the organization's total revenue. First Group is the highest revenue-generating customer, contributing approximately \$9.1 million, followed by First Logistics with \$6.4 million. The remaining customers in the top ten each contribute between \$4.0 million and \$5.4 million, indicating a strong portfolio of high-value customers.

Business Interpretation

The presence of multiple high-revenue customers suggests that the organization has established long-term relationships with several key accounts rather than depending on a single customer. This diversified customer portfolio improves revenue stability and reduces the financial risk associated with losing one major client.

Business Impact

Retaining these high-value customers should be a strategic priority. The company can strengthen customer loyalty by providing dedicated account management, maintaining high service quality, and offering customized logistics solutions. At the same time, understanding the characteristics of these top customers can help the sales team target similar organizations, supporting future revenue growth and customer acquisition.

- **Insight 2: Automotive Freight Generates the Highest Revenue**

The Revenue Distribution by Freight Type visualization shows that Automotive freight is the company's largest revenue-generating segment, contributing approximately \$54.7 million, followed by Food & Beverage (\$48.6 million) and Retail (\$46.1 million). Electronics and Consumer Goods also contribute significantly, while General Freight generates the lowest revenue among the six freight categories.

Business Interpretation

This indicates that the company earns most of its revenue from transporting automotive products. At the same time, the company serves customers across different freight categories, showing that it has a diversified customer base and is not dependent on just one type of freight.

Business Impact

The company should continue focusing on its strong automotive business while maintaining good relationships with customers in this segment. It should also work on increasing business in lower-performing freight categories, such as General Freight, to create a more balanced and stable revenue stream.

- **Insight 3: A Few Transportation Routes Generate the Highest Revenue**

The Top 10 Revenue-Generating Routes chart shows that a small number of transportation routes contribute a significant share of the company's revenue. The route Philadelphia, PA → Seattle, WA generates the highest revenue at approximately \$10.1 million, followed by Charlotte, NC → Portland, OR (\$10.0 million) and Phoenix, AZ → Philadelphia, PA (\$9.8 million). The remaining top routes also generate strong revenue, each contributing between \$7.6 million and \$9.6 million.

Business Interpretation

This indicates that certain routes are more profitable than others, likely due to higher customer demand, larger shipment volumes, or frequent deliveries. These routes play an important role in the company's overall revenue generation.

Business Impact

The company should continue maintaining high service quality and operational efficiency on these profitable routes. It should also study the factors that make these routes successful and apply similar strategies to lower-performing routes to improve overall route profitability.

- **Insight 4: Certain Routes Handle the Highest Shipment Volume**

The Most Frequently Used Routes chart shows that some transportation routes are used much more often than others. The route Columbus, OH → Philadelphia, PA recorded the highest number of loads with 1,628 shipments, followed by Philadelphia, PA → New York, NY (1,537 loads) and Columbus, OH → Los Angeles, CA (1,534 loads). The remaining top routes each handled more than 1,500 shipments, indicating consistently high transportation demand.

Business Interpretation

The high shipment volume on these routes suggests that they are key transportation corridors for the company's operations. These routes experience consistent customer demand and play an important role in maintaining daily logistics activities.

Business Impact

The company should ensure sufficient fleet availability and driver allocation on these high-demand routes to maintain timely deliveries and service quality. Regular monitoring of these routes can also help prevent operational bottlenecks and improve overall transportation efficiency.

- **Insight 5: High Shipment Volume Does Not Always Lead to Higher Revenue**

A comparison of the Top 10 Revenue-Generating Routes and Most Frequently Used Routes reveals that the routes with the highest shipment volume are not necessarily the routes generating the highest revenue. For example, while Columbus, OH → Philadelphia, PA is the most frequently used route, it does not appear among the top revenue-generating routes.

Business Interpretation

This indicates that revenue is influenced not only by the number of shipments but also by factors such as freight type, shipment value, distance, pricing, and customer contracts. Some routes generate higher revenue with fewer loads, while others require a larger number of shipments to achieve similar revenue levels.

Business Impact

Management should evaluate route performance using both **shipment volume** and **revenue** rather than relying on a single metric. This approach will help identify routes that are operationally busy but financially less profitable, allowing the company to optimize pricing strategies, improve route profitability, and make better resource allocation decisions.

- **Insight 6: Revenue is Well Diversified Across Customers and Freight Types**

The dashboard shows that revenue is generated from multiple high-value customers and is distributed across different freight categories. Although First Group is the highest revenue-generating customer and Automotive is the leading freight type, other customers and freight categories also make significant contributions to the company's overall revenue.

Business Interpretation

This indicates that the company is not overly dependent on a single customer or a single freight segment. A diversified customer base and service portfolio reduce business risk and provide greater revenue stability during market fluctuations.

Business Impact

The company should continue expanding its customer base while maintaining a balanced mix of freight services. Diversification will help improve long-term business sustainability and reduce the financial impact of losing any individual customer or industry segment.

- **Insight 7: Customer Value Has a Greater Impact on Revenue Than Shipment Volume**

A comparison of the customer, freight type, and route analyses suggests that revenue is influenced more by the value of customers and freight than by the number of shipments alone. While some routes handle a large number of loads, they do not necessarily generate the highest revenue.

Business Interpretation

This indicates that high-value customers, premium freight categories, and profitable transportation routes contribute more to business performance than shipment volume alone. Revenue growth therefore depends on attracting and retaining profitable customers rather than simply increasing the number of deliveries.

Business Impact

Management should focus on acquiring high-value customers, improving pricing strategies, and expanding profitable freight services. This approach can increase revenue without proportionally increasing operational workload.

3.5 Business Recommendations

Based on the Customer & Route Performance Analysis, the following recommendations are proposed:

- **Strengthen relationships with high-value customers**

The company should focus on retaining its top revenue-generating customers by providing personalized services, dedicated account support, and long-term business agreements. Maintaining strong relationships with these customers will help ensure consistent revenue and improve customer loyalty.

- **Expand high-potential freight categories**

Although Automotive freight generates the highest revenue, the company should also work on increasing business in other freight categories such as Electronics, Consumer Goods, and General Freight. A balanced freight portfolio will reduce business risk and create more stable revenue growth.

- **Optimize routes based on both revenue and shipment volume**

Route planning should consider both profitability and operational demand. High-revenue routes should receive priority to maximize profits, while high-volume routes should be managed efficiently to ensure timely deliveries and better resource utilization.

- **Improve the performance of lower-performing routes**

Management should regularly review routes that generate lower revenue and identify opportunities to improve their performance. Better pricing strategies, route optimization, and attracting new customers in these regions can help increase route profitability.

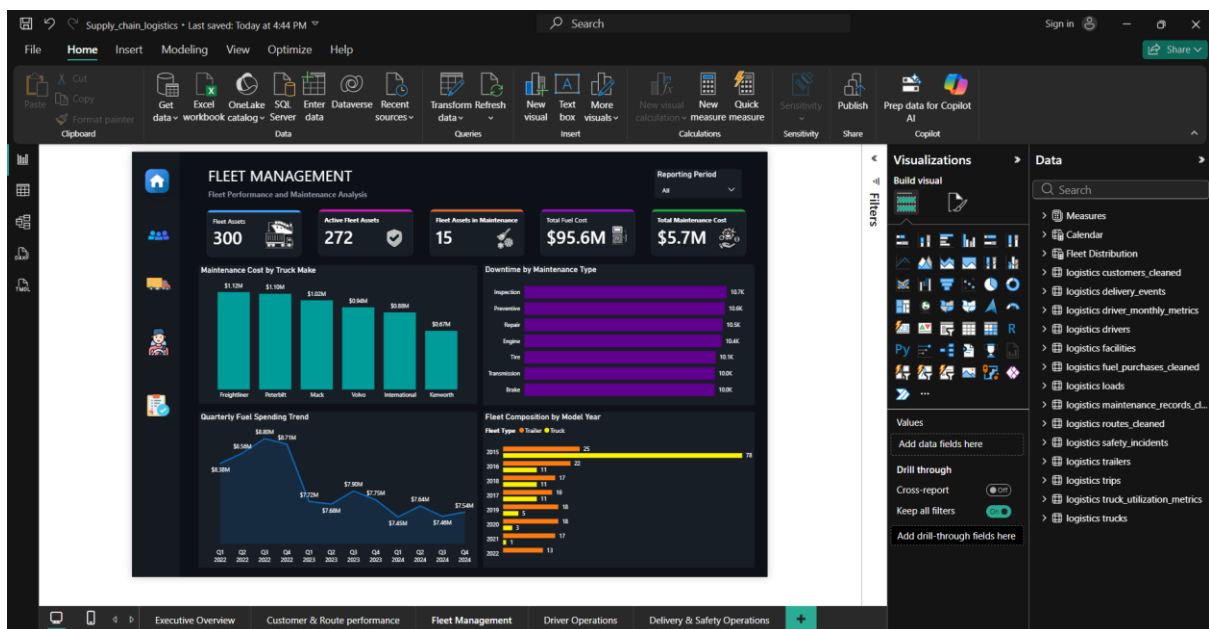
- **Use quarterly performance analysis for better decision-making**

The reporting period filter should be used to monitor customer revenue, freight demand, and route performance over time. Regular analysis of quarterly trends will help management identify seasonal patterns, measure business growth, and make informed strategic decisions.

4. Fleet Management

4.1 Dashboard Objective

The Fleet Management dashboard provides insights into the organization's fleet assets, operating costs, and vehicle utilization. It enables management to monitor fleet size, fuel expenditure, maintenance costs, and vehicle age distribution. By analyzing fleet performance and operational expenses, the dashboard supports informed decision-making related to cost control, asset management, maintenance planning, and fleet optimization.



4.2 Key Performance Indicators (KPIs)



The Fleet Management dashboard provides an overview of the organization's fleet assets and operating expenses. The company currently manages **300 fleet assets**, including trucks and trailers, of which **272 assets are active and available for operations**. In addition, **50 fleet assets are undergoing maintenance**, highlighting the resources allocated for vehicle servicing and repairs.

The dashboard also monitors the major operating expenses associated with fleet management. The **Total Fuel Cost** is **\$95.6 million**, making fuel the largest operational expense, while the **Total Maintenance Cost** amounts to **\$5.7 million**, representing the expenditure incurred to maintain fleet reliability and vehicle performance. These KPIs help management monitor fleet availability, control operational costs, and support effective maintenance planning.

4.3 Visual Analysis

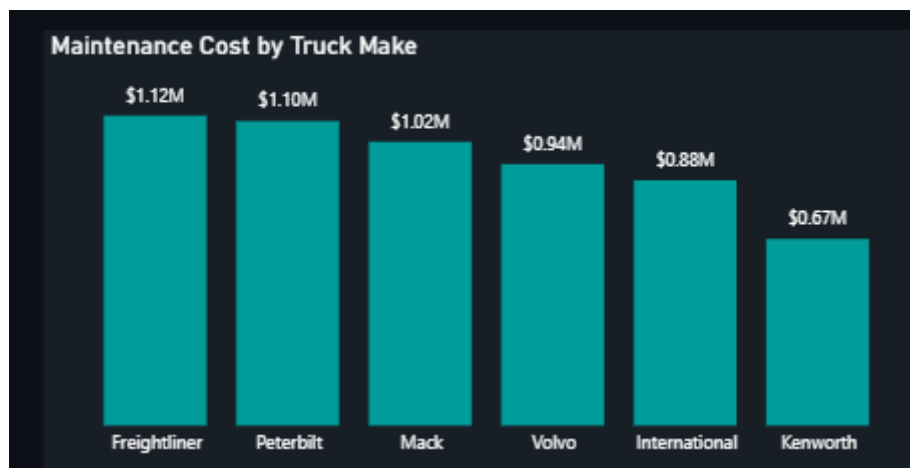
The Customer & Route Performance Analysis dashboard includes four analytical visualizations to evaluate customer contribution and route efficiency:

- **Maintenance cost by truck make:** A vertical bar chart displaying the ten highest revenue-generating customers, enabling management to identify key business accounts and their contribution to overall revenue.

SQL Query : The visualization was created by joining the Trucks table with the Maintenance Records table using the truck_id field. The maintenance cost for each truck make was calculated using the SUM(total_cost) function and grouped by the truck make. The results were then sorted to compare the total maintenance expenditure across different truck manufacturers.

```
111 -- Maintenance Cost by Truck Make
112 v select sum(m.total_cost) as maintenance_cost,
113 t.make
114 from logistics.maintenance_records_cleaned m
115 join logistics.trucks t on t.truck_id = m.truck_id
116 group by t.make
117 order by maintenance_cost desc;
```

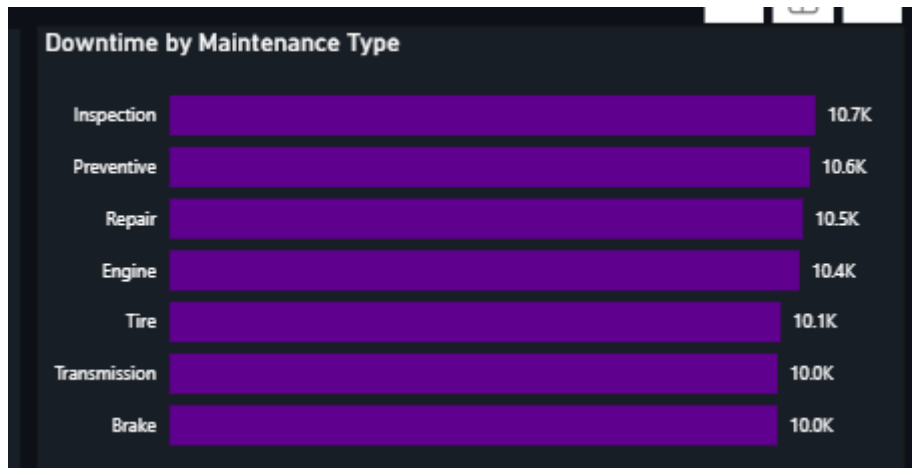
Dashboard Visualization :



- **Downtime by maintenance type**-A horizontal bar chart showing the total downtime hours for each maintenance type, helping management understand which maintenance activities contribute most to fleet downtime

SQL Query : The visualization was created using the Maintenance Records table. Downtime was calculated by summing the total downtime associated with each maintenance type using the SUM(downtime_days) function. The results were grouped by maintenance type to compare the amount of downtime caused by different maintenance activities.

Dashboard Visualization :



- Quarterly Fuel Spending Trend** - A line chart illustrating fuel expenditure across different quarters, allowing management to monitor fuel cost trends over time and evaluate changes in operating expenses.

SQL Query : The visualization was created using the Fuel Purchases table. Fuel costs were aggregated using the SUM(total_cost) function and grouped by year and quarter based on the purchase_date field. The results were then arranged in chronological order to visualize fuel spending trends over time and identify changes in operating costs across different quarters.

```

99
100 -- Quarterly Fuel Spending Trend
101 v select
102   concat('Q', extract(quarter from purchase_date), ' ', extract(year from purchase_date)) as quarter,
103   sum(total_cost) as total_fuel_spend
104 from logistics.fuel_purchases
105 where purchase_date < '2025-01-01'
106 group by extract(year from purchase_date),
107          extract(quarter from purchase_date)
108 order by extract(year from purchase_date),
109          extract(quarter from purchase_date);
110

```

Dashboard Visualization :



- **Fleet Composition by model year** - A clustered column chart comparing the number of trucks and trailers across different model years, enabling management to assess the age distribution of fleet assets and support long-term fleet replacement planning.

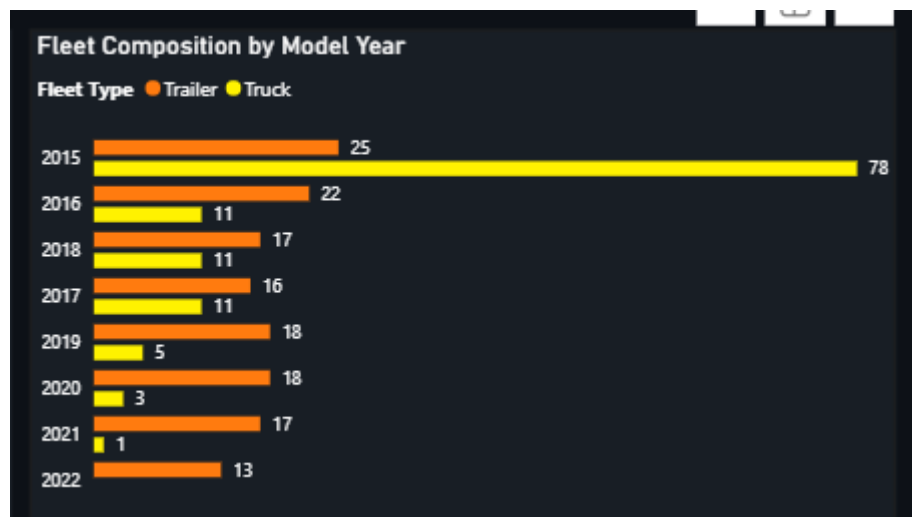
SQL Query : The visualization was created using the Trucks and Trailers tables. The number of trucks and trailers was calculated separately for each model year using the COUNT(*) function. The results from both tables were combined using UNION ALL and grouped by model year to compare the distribution of trucks and trailers across different manufacturing years.

```

117 -- order by maintenance_cost desc,
118
119 -- age distribution
120 v select
121 model_year,
122 sum(truck_count) as total_trucks,
123 sum(trailer_count) as total_trailers
124 from(
125 select model_year,
126 count(*) as truck_count,
127 0 as trailer_count
128 from logistics.trucks
129 group by model_year
130 union all
131 select
132 model_year,
133 0 as truck_count,
134 count(*) as trailer_count
135 from logistics.trailers
136 group by model_year
137 ) as fleet
138 group by model_year
139 order by model_year;

```

Dashboard Visualization :



- **Reporting period Filter** - An interactive Year slicer that allows users to filter all fleet-related visualizations for a selected year, making it easier to analyze fuel spending, maintenance performance, and fleet composition over time.

A separate Calendar table was created in Power BI to provide a common date dimension for the Fleet Management dashboard. The Year field from the Calendar table was used to create the slicer. Relationships were established between the Calendar table and the Fuel Purchases Cleaned table using purchase_date, and between the Calendar table and the Maintenance Records Cleaned table using maintenance_date. This allows a single Year slicer to filter both fuel expenditure and maintenance-related visualizations simultaneously, ensuring consistent time-based analysis across the dashboard.

4.4 Business Insights

- **Insight 1: Freightliner Trucks Incur the Highest Maintenance Cost**

The Maintenance Cost by Truck Make chart shows that Freightliner has the highest maintenance cost at approximately \$1.12 million, followed by Peterbilt (\$1.10 million) and Mack (\$1.02 million). Volvo, International, and Kenworth also contribute significantly to the company's overall maintenance expenditure.

Business Interpretation

The higher maintenance costs associated with Freightliner and Peterbilt trucks may be due to a larger fleet size, higher vehicle utilization, or increased repair requirements. These trucks account for a considerable share of the company's maintenance expenses.

Business Impact

The company should regularly monitor the maintenance performance of these truck makers to identify the causes of higher repair costs. Preventive maintenance schedules, timely inspections, and analysis of recurring repair issues can help reduce maintenance expenses and improve fleet reliability.

- **Insight 2: Downtime is Similar Across Different Maintenance Types**

The Downtime by Maintenance Type chart shows that Inspection has the highest downtime with approximately 10,658 hours, followed closely by Preventive Maintenance (10,570 hours) and Repair (10,453 hours). The other maintenance types, such as Engine, Tire, Transmission, and Brake maintenance, also have similar downtime hours.

Business Interpretation

This shows that vehicle downtime is spread across different types of maintenance instead of being caused by just one type. Routine inspections, preventive maintenance, and repairs all contribute almost equally to the total downtime.

Business Impact

The company should continue improving its overall maintenance planning and scheduling. Reducing downtime across all maintenance activities will help keep more vehicles available for operations, improve fleet efficiency, and reduce delays.

- **Insight 3: Fuel Spending Has Decreased Over Time**

The Quarterly Fuel Spending Trend chart shows that fuel spending was highest during 2022, reaching approximately \$8.8 million in Q3 2022. From 2023 onwards, fuel costs gradually decreased and remained relatively stable, with quarterly spending ranging between \$7.4 million and \$7.9 million throughout 2023 and 2024.

Business Interpretation

The downward trend in fuel spending suggests that the company has been able to better control its fuel expenses over time. This may be due to improved fleet efficiency, optimized route planning, or changes in fuel prices.

Business Impact

The company should continue monitoring fuel consumption and maintain fuel-efficient operating practices. Regular analysis of fuel spending trends can help identify cost-saving opportunities and improve overall fleet operating efficiency.

- **Insight 4: The Fleet Consists of Older Trucks and Newer Trailers**

The Fleet Composition by Model Year chart shows that a large proportion of the company's trucks are 2015 models, with 78 trucks, while the number of trucks decreases significantly in the newer model years. In contrast, trailers are more evenly distributed across different model years, with a larger number of newer trailers in the fleet.

Business Interpretation

This indicates that the truck fleet is relatively older compared to the trailer fleet. Older trucks may require more frequent maintenance and could result in higher operating costs, while the newer trailers suggest that trailer assets have been replaced or expanded more recently.

Business Impact

The company should consider gradually replacing older trucks to improve fuel efficiency, reduce maintenance costs, and increase fleet reliability. Maintaining a balanced fleet with newer vehicles can help improve operational performance and lower long-term operating expenses.

- **Insight 5: Fuel is the Largest Fleet Operating Expense**

The Fleet Management dashboard shows that fuel cost (\$95.6 million) is significantly higher than maintenance cost (\$5.7 million). This indicates that fuel is the company's largest operating expense.

Business Interpretation

Even a small improvement in fuel efficiency can lead to substantial cost savings. Managing fuel consumption therefore has a greater financial impact than reducing maintenance expenses alone.

Business Impact

The company should continue investing in fuel-efficient driving practices, route optimization, and regular vehicle servicing to reduce fuel consumption and improve overall operating profitability.

- **Insight 6: Older Fleet Assets May Increase Maintenance Requirements**

The dashboard shows that a large number of trucks belong to older model years, while maintenance costs remain high across several truck makes. This suggests that the age of the fleet may be contributing to increased maintenance requirements.

Business Interpretation

As vehicles become older, they generally require more frequent servicing and repairs, which can increase maintenance costs and reduce fleet availability.

Business Impact

The company should regularly assess the condition of older trucks and develop a phased fleet replacement plan to improve reliability, reduce maintenance costs, and enhance operational efficiency.

4.5 Business Recommendations

Based on the Fleet Management Analysis, the following recommendations are proposed:

- **Gradually replace older trucks**

The dashboard shows that a large number of trucks belong to older model years. Older vehicles usually require more repairs and may consume more fuel. The company should gradually replace these trucks with newer models to improve fleet reliability, reduce maintenance costs, and increase fuel efficiency.

- **Reduce fuel consumption**

Fuel is the company's largest operating expense. Management should focus on reducing fuel costs by improving route planning, avoiding unnecessary idling, encouraging fuel-efficient driving practices, and ensuring vehicles receive regular servicing. Even a small reduction in fuel consumption can result in significant cost savings.

- **Strengthen preventive maintenance**

The downtime analysis shows that vehicle downtime is spread across different maintenance activities. The company should continue following preventive maintenance schedules and perform regular inspections to identify problems before they become major repairs. This will reduce downtime and keep more vehicles available for operations.

- **Monitor maintenance performance by truck make**

Some truck manufacturers incur higher maintenance costs than others. Management should regularly analyze maintenance records to identify recurring issues, compare the performance of different truck makes, and take corrective actions where necessary. This will help reduce repair costs and improve fleet performance.

- **Use fleet data for long-term planning**

The Fleet Management dashboard should be reviewed regularly to monitor fuel costs, maintenance expenses, and the age of fleet assets. Using this information for planning will help management make better decisions about fleet expansion, vehicle replacement, budgeting, and overall operational efficiency.

5. Driver Operations

5.1 Dashboard Objective

The Driver Operations dashboard monitors and evaluates driver performance across the logistics network. It provides a centralized view of key operational metrics that help managers assess driver productivity, revenue contribution, fuel efficiency, trip completion, and delivery performance, enabling decision-makers to identify high-performing drivers, monitor operational trends, and recognize areas that require improvement. The dashboard also includes a Home Terminal filter that allows users to analyze driver performance across different regional locations.

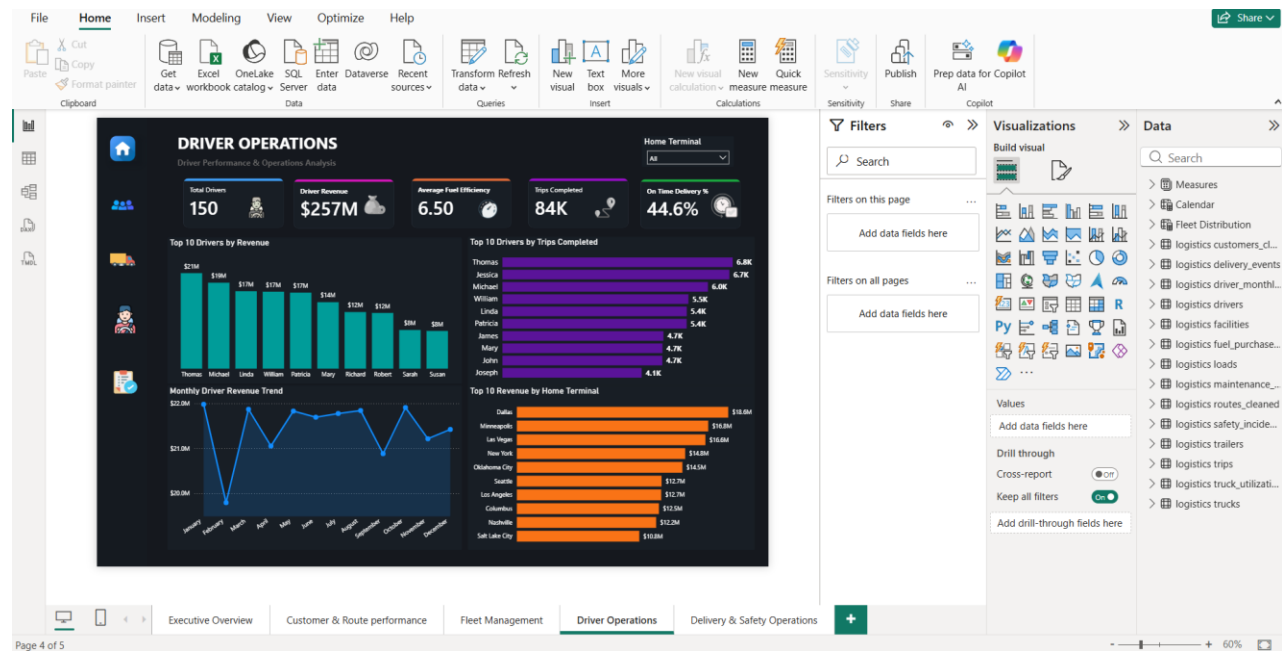


Figure 5.1 Driver Operations Dashboard – Overview

5.2 Key Performance Indicators (KPIs)

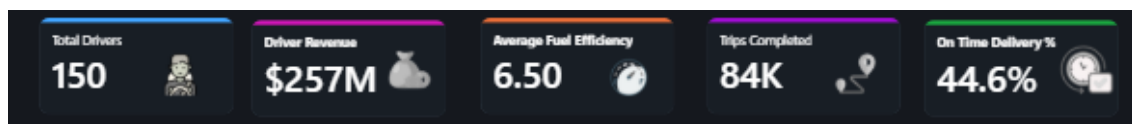


Figure 4.2 Key Performance Indicator (KPI) Summary Strip

The Driver Operations dashboard consists of five key performance indicators that provide an overview of workforce scale, revenue contribution, and operational efficiency. The company operates a workforce of **150 drivers**, who together completed **84,000 trips** during the reporting period, reflecting the scale of daily transportation activity.

The dashboard also highlights three additional performance metrics. **Driver Revenue** stands at **\$257 million**, representing the total revenue generated by all drivers. **Average Fuel Efficiency** is **6.50 MPG**, reflecting fuel consumption across the fleet. Additionally, **On-Time Delivery Percentage** stands at **44.6%**, indicating the share of deliveries completed within the scheduled window. Together, these KPIs help management evaluate workforce productivity, financial contribution, and service reliability.

5.3 Visual Analysis

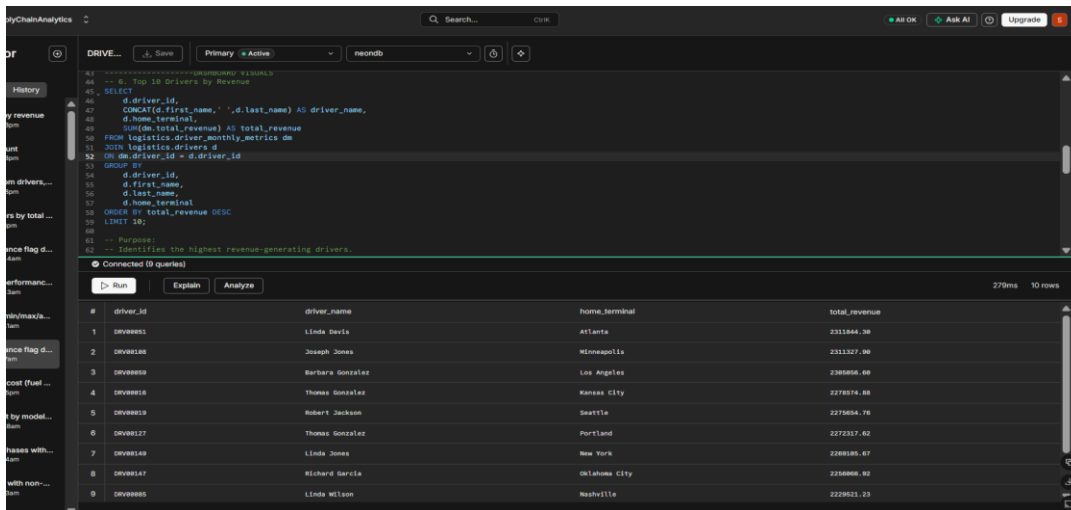
The Driver Operations dashboard includes four analytical visualizations to evaluate driver productivity and revenue performance. Each visualization below is presented together with its underlying SQL query, the resulting dashboard chart, and a short explanation of the business insight it provides.

Visualization 1: Top 10 Drivers by Revenue

A vertical bar chart displaying the ten highest revenue-generating drivers, enabling management to identify top performers and their contribution to overall business revenue.

SQL Query

The visualization was developed by joining the drivers and driver_monthly_metrics tables using the driver_id field. Revenue was aggregated for each driver using the SUM(revenue) function, after which the results were sorted in descending order and the top ten drivers were selected.



```
43 --===== Unnamed Visuals =====
44 --> Top 10 Drivers by Revenue
45 SELECT
46     d.driver_id,
47     CONCAT(d.first_name, ' ', d.last_name) AS driver_name,
48     d.home_terminal,
49     SUM(dm.total_revenue) AS total_revenue
50 FROM logistics.driver_monthly_metrics dm
51 JOIN logistics.drivers d
52 ON dm.driver_id = d.driver_id
53 GROUP BY
54     d.driver_id,
55     d.first_name,
56     d.last_name,
57     d.home_terminal
58 ORDER BY total_revenue DESC
59 LIMIT 10;
60
61 -- Purpose:
62 -- Identifies the highest revenue-generating drivers.
```

#	driver_id	driver_name	home_terminal	total_revenue
1	DRV0001	Linda Davis	Atlanta	231844.38
2	DRV0188	Joseph Jones	Minneapolis	231327.98
3	DRV0059	Barbara Gonzalez	Los Angeles	228906.08
4	DRV0015	Thomas Gonzalez	Kansas City	227874.08
5	DRV0019	Robert Jackson	Seattle	227684.76
6	DRV0127	Thomas Gonzalez	Portland	227317.82
7	DRV0149	Linda Jones	New York	226185.07
8	DRV0147	Richard Garcia	Oklahoma City	226088.92
9	DRV0005	Linda Wilson	Nashville	225921.23

Figure 5.3 SQL Query – Top 10 Drivers by Revenue

Dashboard Visualization

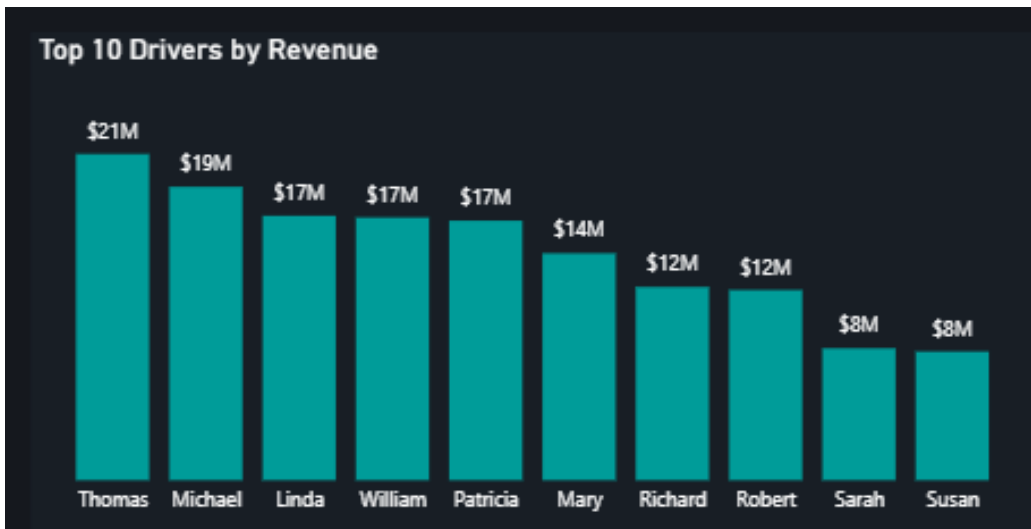


Figure 5.4 Dashboard Visualization – Top 10 Drivers by Revenue

The chart shows that revenue is concentrated among a small group of drivers, with the top performer generating noticeably higher revenue than the rest of the group. This highlights a clear performance gap between the leading drivers and the remaining workforce.

Visualization 2: Top 10 Drivers by Trips Completed

A bar chart ranking drivers by the total number of completed trips, highlighting operational productivity rather than financial contribution.

SQL Query

This visualization was developed by joining the drivers and driver_monthly_metrics tables using the driver_id field. The total number of completed trips for each driver was calculated using the SUM(trips_completed) function. The results were grouped by driver name, sorted in descending order based on total trips completed, and the top ten drivers were selected for visualization.

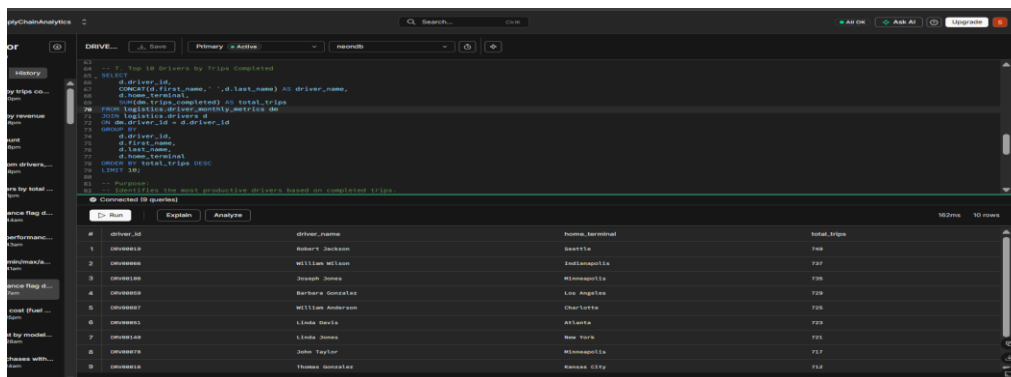


Figure 5.5 SQL Query – Top 10 Drivers by Trips Completed

Dashboard Visualization

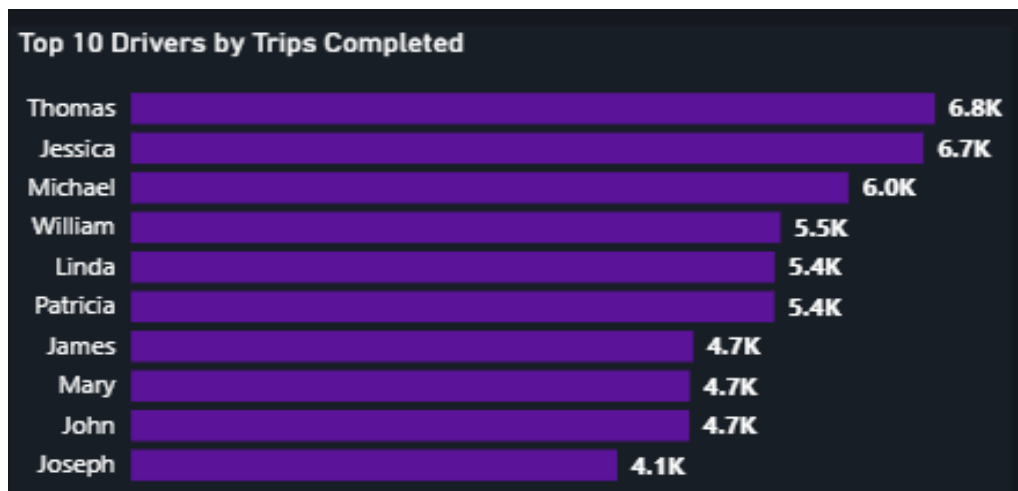


Figure 5.6 Dashboard Visualization – Top 10 Drivers by Trips Completed

The ranking of drivers by trips completed differs from the ranking by revenue, showing that the highest trip volume does not always translate into the highest earnings. This indicates that route value and freight type influence revenue more than the number of trips alone.

Visualization 3: Monthly Driver Revenue Trend

A line chart illustrating revenue generated by drivers over time, helping management track monthly fluctuations and evaluate financial stability.

SQL Query

This query was developed by grouping revenue records from the driver_monthly_metrics table according to the reporting month, and aggregating revenue using the SUM(revenue) function to generate the monthly trend.

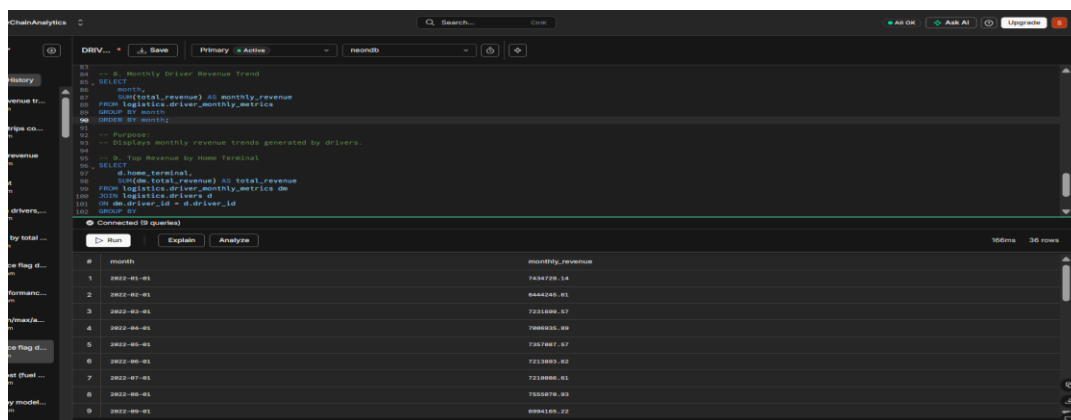


Figure 5.7 SQL Query – Monthly Driver Revenue Trend

Dashboard Visualization

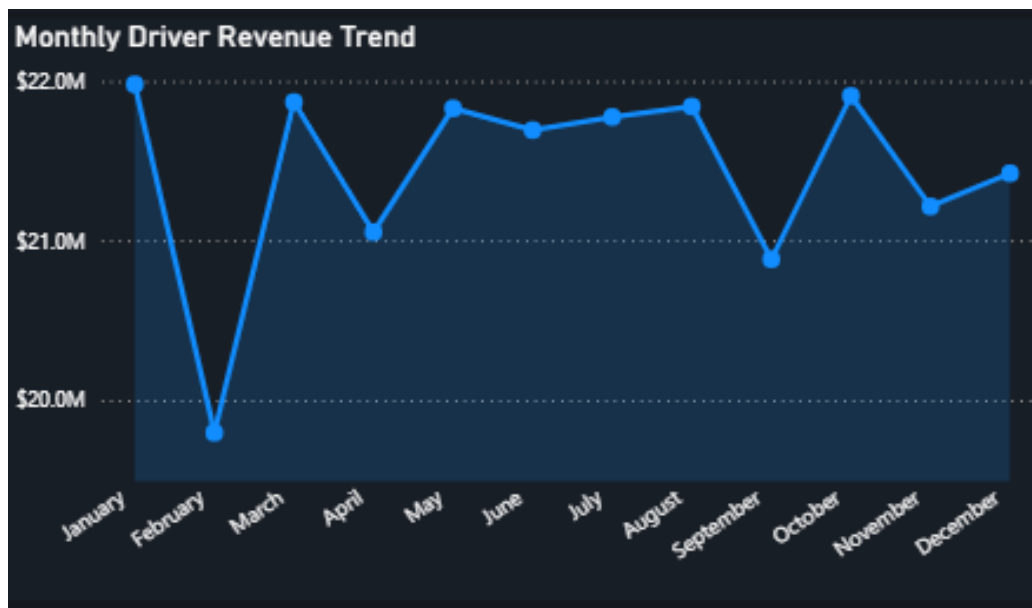


Figure 5.8 Dashboard Visualization – Monthly Driver Revenue Trend

The trend line shows relatively stable monthly revenue with only moderate fluctuations across the reporting period. This consistency suggests that transportation operations are running smoothly and workload is fairly balanced from month to month.

Visualization 4: Top Revenue by Home Terminal

A horizontal bar chart comparing total revenue generated by drivers across different home terminals, supporting regional performance evaluation.

SQL Query

This visualization was created by joining the drivers and driver_monthly_metrics tables and aggregating revenue using the SUM(revenue) function, grouped by each driver's assigned home_terminal field.

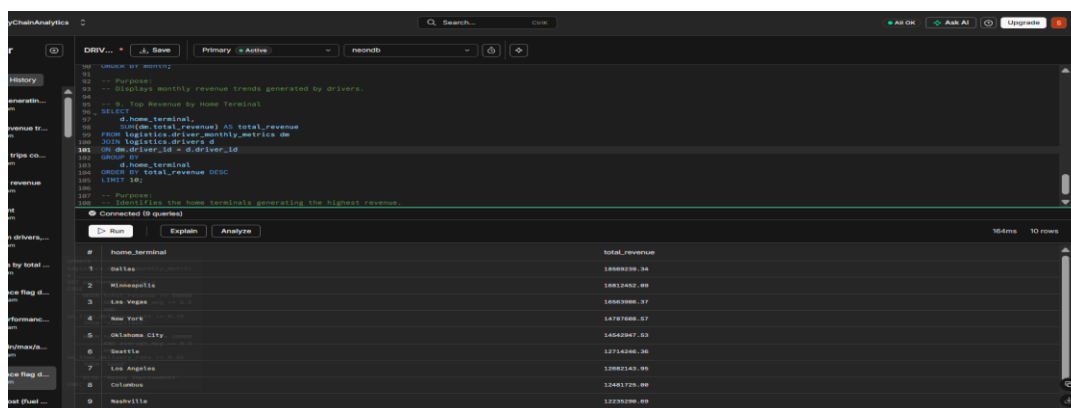


Figure 5.9 SQL Query – Top Revenue by Home Terminal

Dashboard Visualization

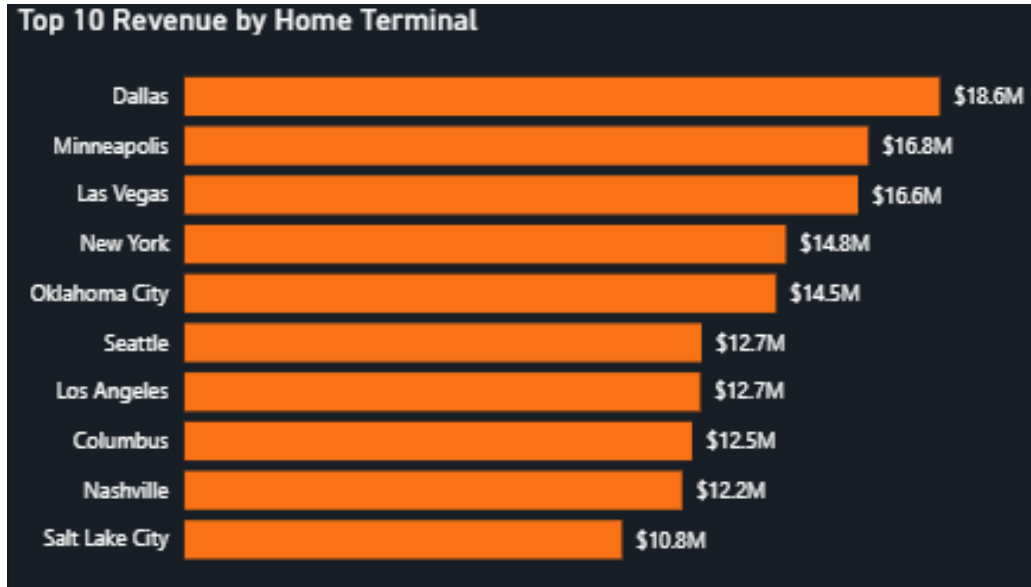


Figure 5.10 Dashboard Visualization – Top Revenue by Home Terminal

Certain home terminals generate noticeably higher revenue than others, pointing to differences in regional demand, route allocation, or driver experience across locations. This comparison helps management identify terminals that could benefit from best-practice sharing.

Home Terminal Filter

A Home Terminal slicer has been included to enable users to analyze driver revenue and trip performance filtered by specific terminals. This interactive filter allows management to compare performance across regions and identify terminal-specific trends that support workforce and resource planning.

5.4 Business Insights

Insight 1: Revenue Generation is Concentrated Among Top Drivers

The Top 10 Drivers by Revenue visualization shows that a relatively small group of drivers contributes a significant share of total driver revenue, indicating that revenue generation is not evenly distributed across the workforce.

Business Interpretation

High-performing drivers may be assigned more profitable routes, possess greater operational experience, or demonstrate stronger customer service and delivery efficiency.

Business Impact

Management should identify the practices followed by these top-performing drivers and incorporate them into training programs. Recognizing and rewarding these employees can also improve motivation and overall workforce performance.

Insight 2: High Trip Volume Does Not Always Result in Higher Revenue

A comparison between the Top Drivers by Revenue and Top Drivers by Trips Completed shows that drivers completing the highest number of trips are not necessarily the highest revenue generators.

Business Interpretation

Some routes generate significantly higher revenue despite involving fewer trips, suggesting that route profitability and freight value influence revenue more than trip volume alone.

Business Impact

Business decisions should weigh route profitability and shipment value, not just driver productivity, to maximize operational returns.

Insight 3: Monthly Revenue Shows Stable Operational Performance

The Monthly Driver Revenue Trend shows relatively consistent revenue generation throughout the reporting period, with only moderate fluctuations.

Business Interpretation

Stable monthly revenue indicates that transportation operations are functioning consistently and the organization has maintained a balanced workload across months.

Business Impact

Consistent revenue improves financial forecasting, budgeting, and long-term operational planning while reducing uncertainty.

Insight 4: Certain Home Terminals Outperform Others

The Top Revenue by Home Terminal visualization highlights that some terminals contribute significantly more revenue than others.

Business Interpretation

Higher-performing terminals may benefit from stronger customer demand, better route allocation, more experienced drivers, or improved operational efficiency.

Business Impact

Management should study the operational strategies of high-performing terminals and apply similar practices across lower-performing locations.

Insight 5: Driver Productivity Can Be Evaluated from Multiple Perspectives

The dashboard combines financial and operational metrics, allowing driver performance to be evaluated using both revenue generation and completed trips.

Business Interpretation

Evaluating multiple KPIs provides a more balanced assessment of driver performance than relying on a single metric.

Business Impact

This multidimensional evaluation supports fair performance reviews, incentive programs, and workforce planning.

Insight 6: Fuel Efficiency Directly Influences Operating Costs

The dashboard reports an average fuel efficiency of approximately 6.50 MPG, providing an overview of fleet fuel utilization.

Business Interpretation

Fuel efficiency is a key operational metric because fuel expenses represent one of the largest costs in logistics operations.

Business Impact

Monitoring fuel efficiency helps reduce operational costs, improve sustainability, and identify drivers who may benefit from eco-driving training.

Insight 7: On-Time Delivery Performance Requires Improvement

The dashboard indicates an overall On-Time Delivery Rate of 44.6%, suggesting that a large share of deliveries are not completed within the expected schedule.

Business Interpretation

Delivery delays may result from inefficient route planning, traffic conditions, scheduling issues, or operational bottlenecks.

Business Impact

Improving on-time delivery performance can increase customer satisfaction, strengthen service reliability, and enhance the company's competitive position.

Insight 8: Integrated Driver Analytics Improve Operational Decision-Making

By combining KPIs, trend analysis, regional comparisons, and driver rankings, the dashboard provides a comprehensive view of driver operations.

Business Interpretation

Managers can evaluate workforce performance from both strategic and operational perspectives without relying on multiple reports.

Business Impact

A centralized dashboard improves decision-making speed, increases reporting efficiency, and supports data-driven operational management across the logistics network.

5.5 Business Recommendations

Based on the Driver Operations dashboard analysis, the following recommendations are proposed:

1. **Introduce performance-based incentive programs:** Reward drivers who consistently generate high revenue, maintain excellent delivery performance, and demonstrate efficient operational practices. Recognition programs can improve employee motivation and encourage continuous performance improvement.
2. **Improve driver training programs:** Analyze the working practices of top-performing drivers and incorporate them into structured training programs for newly recruited and average-performing drivers. This can improve overall workforce productivity and operational consistency.
3. **Optimize route allocation:** Allocate high-value routes based on driver experience, historical performance, and delivery efficiency. Optimized route assignment can improve both revenue generation and customer satisfaction.
4. **Focus on improving on-time delivery performance:** Investigate the primary causes of delayed deliveries and implement corrective measures such as route optimization, better scheduling, real-time traffic monitoring, and improved dispatch coordination.
5. **Enhance fuel efficiency monitoring:** Regularly monitor driver fuel efficiency and promote eco-driving practices through awareness programs and performance tracking. Improved fuel efficiency will reduce transportation costs and support sustainable logistics operations.

6. **Benchmark high-performing home terminals:** Conduct detailed performance evaluations of the highest revenue-generating terminals and implement their successful operational strategies across other regional locations to improve organizational performance.
7. **Continue data-driven performance monitoring:** Use the Driver Operations dashboard as a continuous performance monitoring tool to track KPIs, identify operational issues early, and support informed management decisions. Regular dashboard reviews will help maintain operational efficiency and drive continuous improvement.

6. Delivery & Safety Operations

6.1 Dashboard Objective

The Delivery & Safety Operations dashboard provides a consolidated view of delivery performance, route efficiency, and transportation safety across the logistics network. It brings together the total number of deliveries, loads, routes, and safety incidents into a single interactive dashboard, enabling managers to evaluate operational activity from one interface rather than multiple disconnected reports.

Transportation companies process large volumes of deliveries every month while simultaneously managing route efficiency and safety risk. Without a centralized analytical view, identifying delayed deliveries, underperforming routes, and recurring safety incidents becomes difficult and time-consuming, which can delay corrective action and increase operational cost. This dashboard addresses that gap by transforming delivery and safety data into actionable business insights, supporting faster and more informed operational decisions.

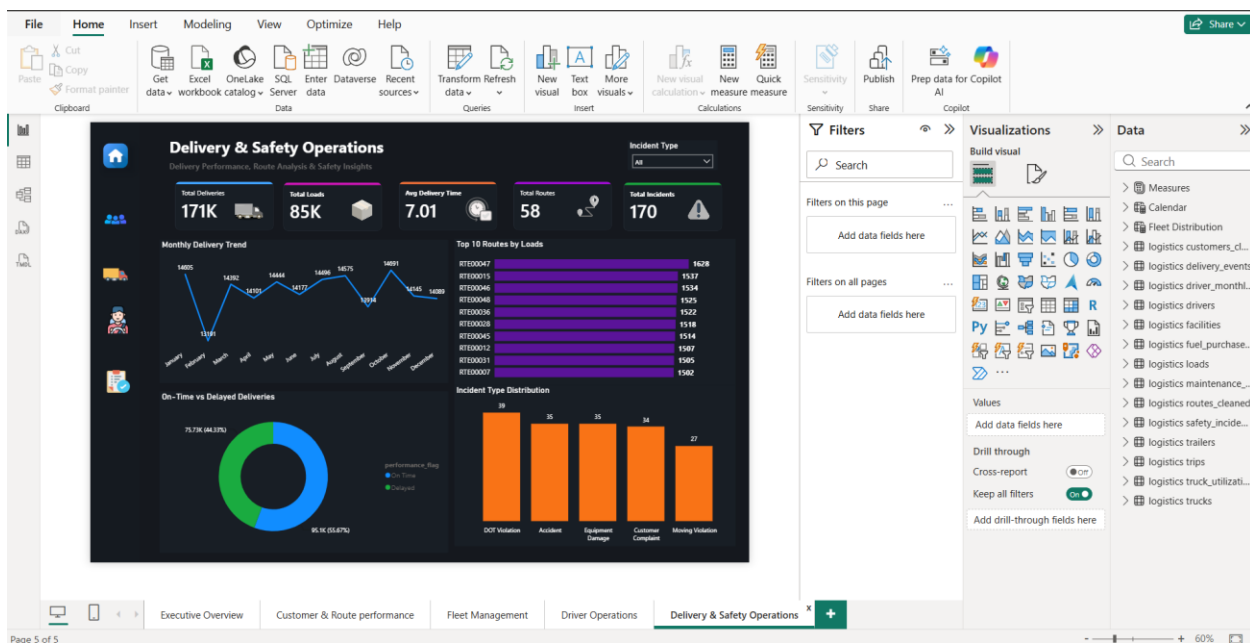


Figure 6.1 Delivery & Safety Operations Dashboard – Overview

6.2 Key Performance Indicators (KPIs)

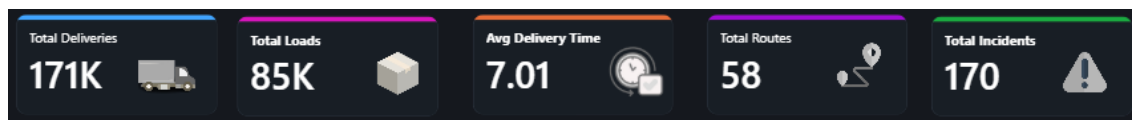


Figure 6.2 Key Performance Indicator (KPI) Summary Strip

The Delivery & Safety Operations dashboard consists of five key performance indicators that summarize the overall scale and reliability of transportation operations. During the reporting

period, the network completed 171K total deliveries across 85K loads, reflecting a high-volume, large-scale logistics operation.

The dashboard also highlights three additional operational metrics. **Average Delivery Time** stands at **7.01**, reflecting the typical duration required to complete a delivery cycle. The network is supported by **58 total routes**, indicating broad geographical coverage, while **170 recorded safety incidents** highlight an area that requires continued monitoring. Together, these KPIs give management an immediate view of delivery scale, efficiency, and operational risk.

6.3 Visual Analysis

The Delivery & Safety Operations dashboard includes four analytical visualizations that evaluate delivery trends, route utilization, on-time performance, and safety incidents. Each visualization below is presented together with a summary of its underlying SQL logic, the resulting dashboard chart, and a short explanation of the business insight it provides.

Visualization 1: Monthly Delivery Trend

A line chart illustrating the total number of deliveries completed each month, allowing management to monitor fluctuations in transportation activity over time.

SQL Query

The query aggregates records from the `delivery_events` table by the month of completion using `COUNT()`, producing a monthly time series of delivery volume.

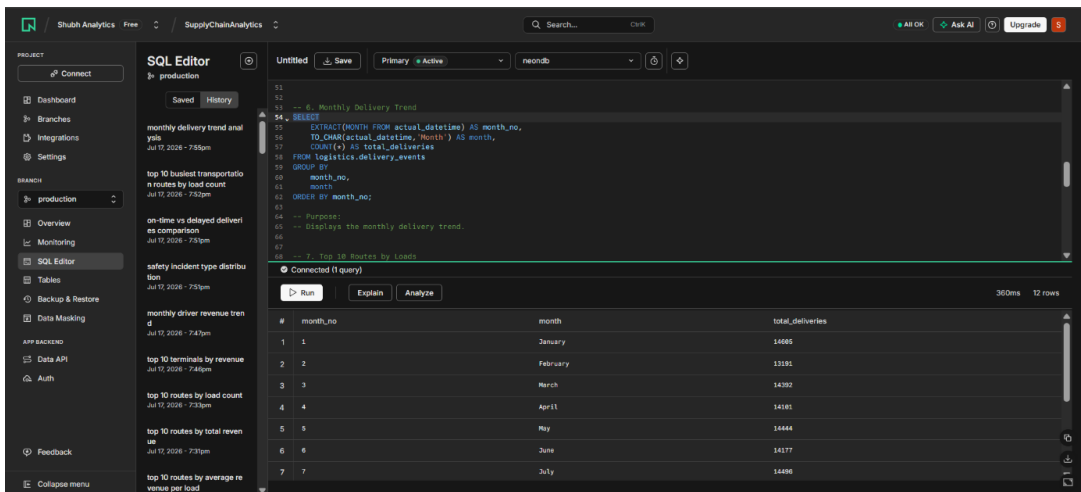


Figure 6.3 SQL Query – Monthly Delivery Trend

Dashboard Visualization

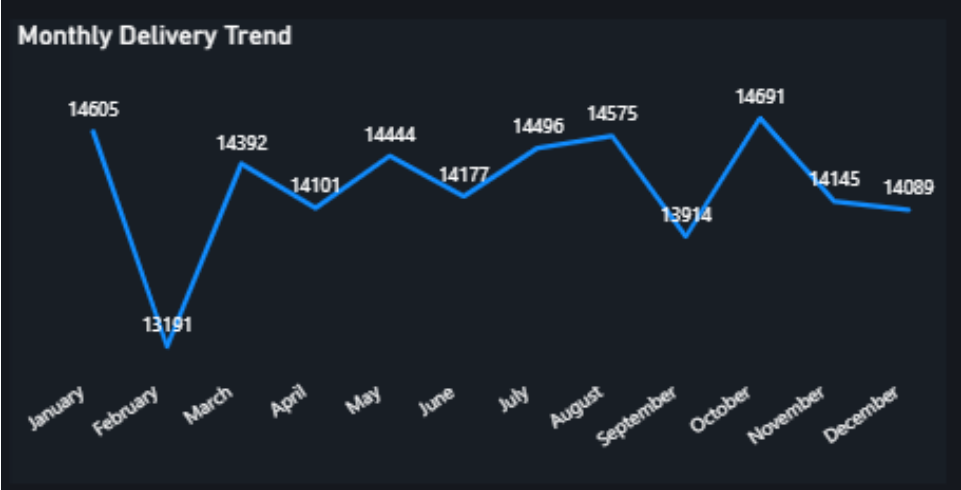


Figure 6.4 Dashboard Visualization – Monthly Delivery Trend

Delivery volumes remain relatively stable from month to month, with only moderate fluctuations. This consistency suggests that transportation operations are well-balanced and workload is distributed evenly across the reporting period.

Visualization 2: Top 10 Routes by Loads

A horizontal bar chart ranking transportation routes by the total number of loads handled, highlighting the busiest corridors in the network.

SQL Query

This visualization joins the routes and loads tables on route_id, aggregates load counts using COUNT(), and sorts the results in descending order to surface the ten highest-volume routes.

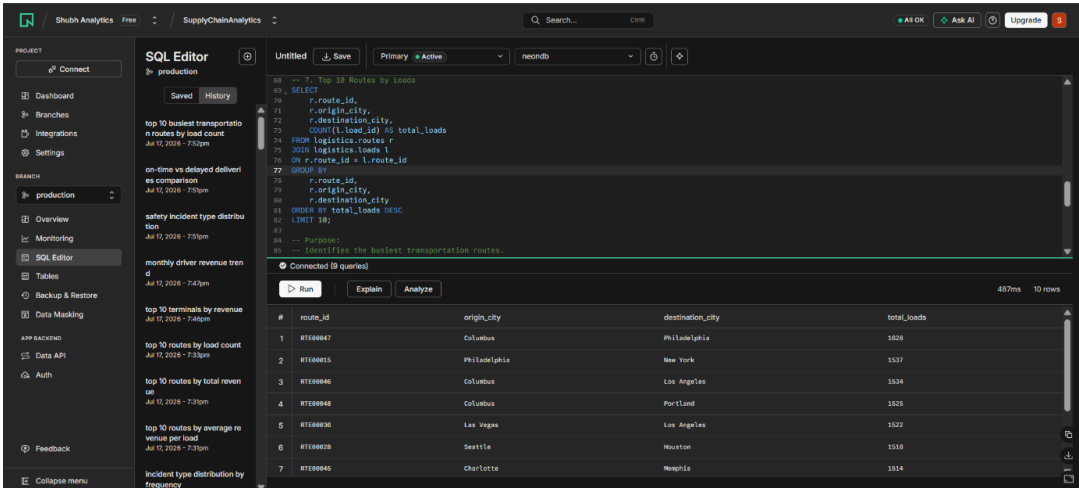


Figure 6.5 SQL Query – Top 10 Routes by Loads

Dashboard Visualization



Figure 6.6 Dashboard Visualization – Top 10 Routes by Loads

A small number of routes carry a disproportionately high share of total loads, indicating that freight demand is concentrated on a limited set of high-value corridors rather than spread evenly across the network.

Visualization 3: On-Time vs Delayed Deliveries

A donut chart comparing deliveries completed within the scheduled window against those that experienced delays, offering an immediate view of delivery reliability.

SQL Query

This query groups delivery_events by delivery status and counts the number of on-time and delayed records using COUNT(), producing the two-way split shown on the dashboard.

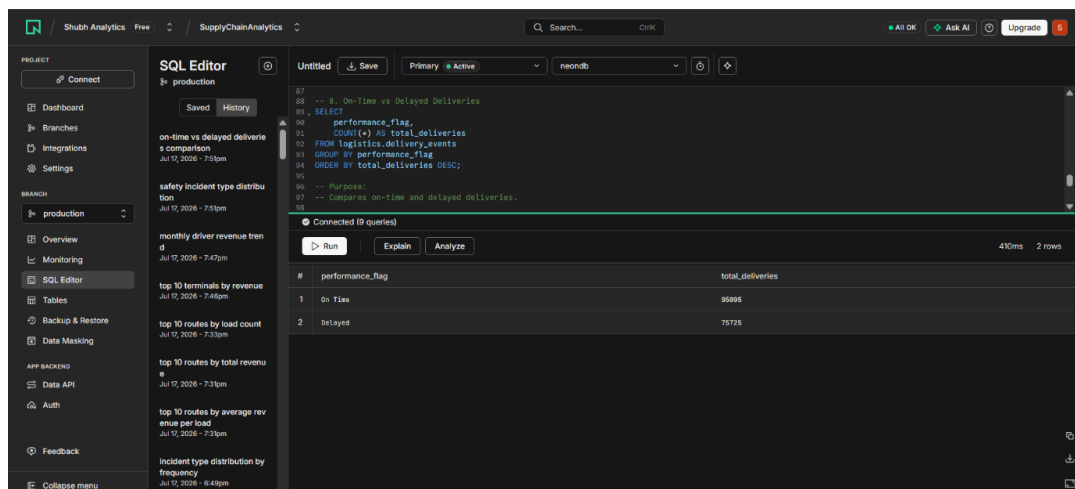


Figure 6.7 SQL Query – On-Time vs Delayed Deliveries

Dashboard Visualization

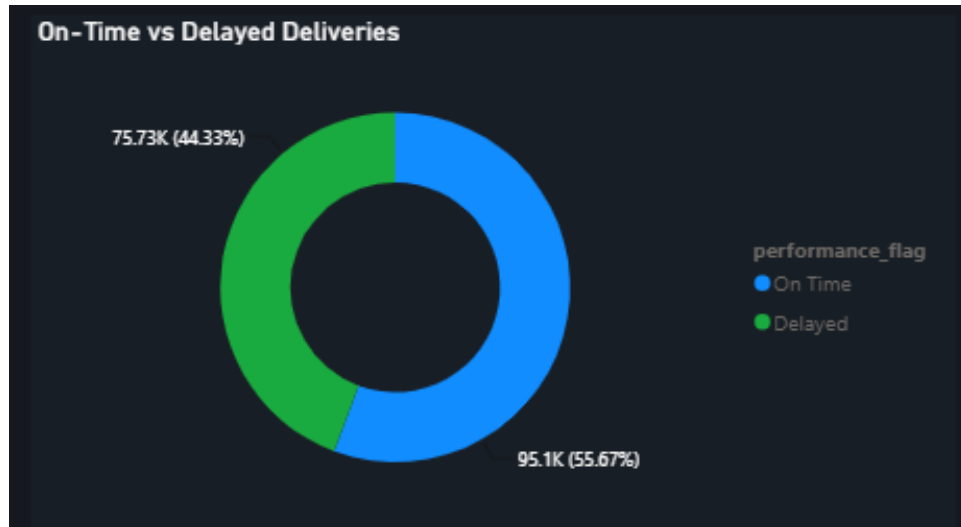


Figure 6.8 Dashboard Visualization – On-Time vs Delayed Deliveries

While most deliveries are completed on time, a meaningful share continue to experience delays. This gap represents a clear opportunity to improve scheduling, routing, and dispatch coordination.

Visualization 4: Incident Type Distribution

A column chart displaying the frequency of different safety incident categories, including accidents, equipment damage, customer complaints, moving violations, and DOT violations.

SQL Query

This query groups records from the `safety_incidents` table by `incident_type` and counts occurrences using `COUNT()`, ranking categories from most to least frequent.

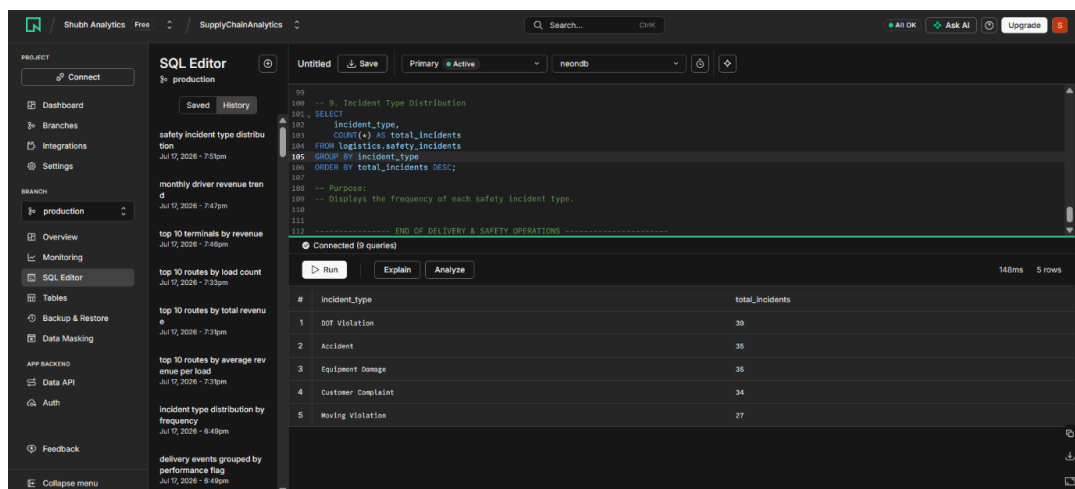


Figure 6.9 SQL Query – Incident Type Distribution

Dashboard Visualization

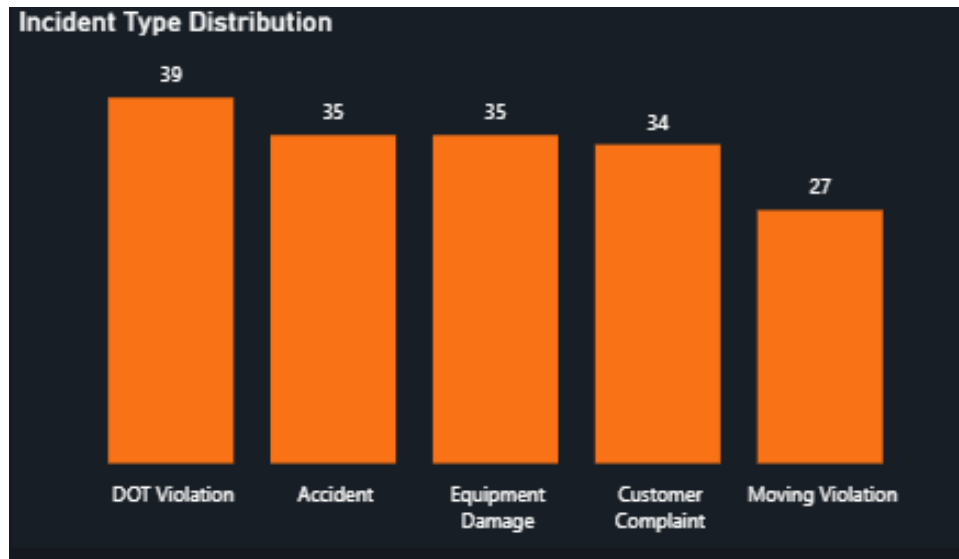


Figure 5

6.10 Dashboard Visualization – Incident Type Distribution

DOT violations occur more frequently than any other incident category, followed by accidents and equipment damage. This points to compliance-related gaps, such as driver documentation or vehicle inspection lapses, as the leading source of operational risk.

6.4 Business Insights

The Delivery & Safety Operations dashboard provides valuable insight into delivery performance, route utilization, and transportation safety. The following insights were derived from the dashboard analysis.

Insight 1: Delivery Operations Maintain a High Operational Volume

The organization completed approximately 171,000 delivery events during the reporting period, reflecting a large-scale, consistently active transportation operation.

Business Interpretation

A high delivery volume demonstrates strong operational capacity and the organization's ability to manage large-scale transportation activity.

Business Impact

Continuous monitoring of delivery volume enables management to plan fleet capacity, workforce allocation, and infrastructure investment while maintaining service quality.

Insight 2: Transportation Demand is Concentrated on Selected Routes

The Top 10 Routes by Loads visualization shows that a limited number of routes handle a significantly higher share of loads than the rest of the network.

Business Interpretation

Certain routes experience consistently higher freight demand due to customer concentration, regional distribution centers, or supply chain requirements.

Business Impact

High-demand routes should receive priority in fleet allocation, maintenance scheduling, and driver assignment to avoid operational bottlenecks.

Insight 3: On-Time Delivery Performance Has Room for Improvement

Although most deliveries are completed on time, a considerable proportion of shipments continue to experience delays.

Business Interpretation

Delivery delays may result from traffic congestion, inefficient route planning, vehicle breakdowns, loading delays, or scheduling inefficiencies.

Business Impact

Reducing delayed deliveries can meaningfully improve customer satisfaction, strengthen service reliability, and lower the operational cost of missed delivery windows.

Insight 4: DOT Violations Represent the Most Frequent Safety Issue

The Incident Type Distribution visualization shows that DOT violations occur more often than accidents, equipment damage, customer complaints, or moving violations.

Business Interpretation

Frequent DOT violations may point to compliance-related challenges involving driver documentation, vehicle inspections, or operating-hour regulations.

Business Impact

Reducing DOT violations improves regulatory compliance, minimizes legal exposure, and strengthens the organization's operational reputation.

Insight 5: Centralized Monitoring Improves Operational Visibility

By integrating delivery, route, and safety data into a single dashboard, management gains a comprehensive view of transportation performance without consulting multiple reports.

Business Interpretation

Decision-makers can quickly identify operational issues, evaluate performance trends, and investigate safety concerns from one interface.

Business Impact

Centralized reporting improves the speed of decision-making and supports proactive operational management across the logistics network.

6.5 Business Recommendations

Based on the findings from the Delivery & Safety Operations dashboard, the following recommendations are proposed to improve delivery efficiency, transportation safety, and overall logistics performance:

1. Improve on-time delivery performance: Implement route optimization, GPS-based navigation, and real-time traffic monitoring to reduce delivery delays and improve schedule adherence.
2. Optimize high-demand routes: Allocate additional vehicles, drivers, and resources to consistently high-volume routes, and periodically re-evaluate route utilization to reduce congestion.
3. Strengthen safety compliance programs: Conduct regular safety audits and driver compliance checks to reduce DOT violations and improve regulatory adherence.
4. Enhance driver safety training: Provide periodic training on defensive driving, accident prevention, and emergency response to reduce incident rates.
5. Implement predictive maintenance: Schedule preventive maintenance for trucks and trailers to reduce breakdowns and minimize delivery disruptions.
6. Continue data-driven monitoring: Use the Delivery & Safety Operations dashboard as an ongoing performance tool to track KPIs and support continuous operational improvement.

7. Conclusion

7.1 Project Summary

This project set out to build a complete, end-to-end analytics solution for a logistics and supply chain business, taking raw operational data all the way through to interactive, decision-ready dashboards. Data covering customers, routes, fleet assets, drivers, fuel purchases, maintenance records, deliveries, and safety incidents was consolidated into a structured PostgreSQL database and analyzed using SQL. The resulting insights were then translated into six interactive Power BI dashboards - Executive Summary, Customer & Route Performance, Fleet Management, Driver Operations, and Delivery & Safety Operations - each designed to answer specific business questions for a different stakeholder audience, from senior management to operations teams.

Across the project, the emphasis was on connecting technical work - database design, SQL querying, and DAX-driven calculations - directly to business outcomes such as revenue concentration, fleet reliability, driver productivity, and delivery performance. This approach demonstrates how raw operational data can be turned into a single source of truth that supports faster, more confident decision-making.

7.2 Objectives Achieved

- Successfully integrated multiple logistics datasets into PostgreSQL.
- Developed an optimized relational database.
- Created interactive Power BI dashboards.
- Generated business insights through SQL and visualization.

7.3 Major Business Findings

- High-value customers contribute a significant portion of revenue.
- Fleet utilization remains efficient with opportunities for maintenance optimization.
- Driver performance varies across revenue and trip completion.
- Delivery operations are generally stable, while safety monitoring remains important.
- Executive dashboard provides a consolidated organizational view.

7.4 Technologies Used

- PostgreSQL for database management.
- SQL for querying and data aggregation.

- Power BI for interactive dashboards.
- DAX for KPI calculations.
- Power Query for data preparation.

7.5 Learning Outcomes

- Database design.
- SQL joins.
- Aggregation.
- Data modeling.
- Dashboard design.
- Business analytics.
- Data storytelling.

7.6 Final Remarks

Overall, this project demonstrates how data analytics can improve logistics decision-making by combining database management, SQL analysis, and interactive business intelligence dashboards. By moving from raw, fragmented operational data to a unified relational database and then to a set of purpose-built dashboards, the project shows a practical, end-to-end workflow that mirrors how real-world logistics and supply chain organizations use data to monitor performance, control costs, and plan strategically. It reflects the growing importance of business intelligence tools in transforming everyday operational data into a genuine competitive advantage.

